



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Department of Computer Science and Engineering (Data Science)

Polycystic Kidney Disease (PKD) Detection System

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Agenda

- **Introduction**
- **Literature Review**
- **Research Gap**
- **Problem Definition**
- **Proposed Design**

Introduction

- **Polycystic Kidney Disease (PKD)** is a **genetic disorder** where **many fluid-filled cysts** grow in the kidneys.
- These cysts cause the **kidneys to enlarge**, reducing their **function** and sometimes leading to **kidney failure**.
- **Traditional diagnosis** depends on **manual MRI/CT image analysis** and **genetic testing**, which is **slow** and **time-consuming**.
- Such methods are **subjective** and often **detect the disease in late stages**, missing early warning signs.
- Using **Machine Learning (ML)**, we can **automate cyst detection**, **classify disease severity**, and **predict disease progression**.
- Our project builds an **ML-Enhanced PKD Detection System** that provides **early, accurate, and explainable diagnosis**, helping **doctors make better decisions** and **improving patient outcomes**.

Literature Review

Title	Author	Methodology	Gaps
Deep Learning-Based Automated Imaging Classification of ADPKD	Youngwoo Kim, Seonah Bu, Cheng Tao, Kyongtae T. Bae	Used Transfer Learning with CNNs and Explainable AI (XAI) to classify typical and atypical PKD from MRI images.	Limited to image classification; no integration with genetic or clinical data; lower confidence in atypical cases.
Deep Learning for PKD: Utilizing Neural Networks for Early Detection through Gene Expression	Kapil Panda, Anirudh Mazumder	Applied MLPs with backpropagation , K-Means Clustering , and Stacking Ensembles (SVC, RF, GB, LR) on gene expression data.	Limited real-world gene data; trained mostly on animal datasets; lacks explainability (no XAI).
An AI-Generated Algorithm to Measure Total Kidney Volume in ADPKD	Taylor, J., Thomas, R., Metherall	Used Ensemble U-Net (nnU-Net) for kidney segmentation and volume measurement from MRI.	Dataset bias (young, PKD1-dominant); limited to specific scanners; no real-time clinical deployment.
Characterizing the ADPKD-IFT140 Phenotypic Signature with Deep Learning	Ahmad Ghannem, Fadi George Munairdjy	Used Deep Learning for Cystic Phenotypic Signature Analysis (CPSA) and cystic index calculation.	Model focused on one genotype (IFT140); lacks generalization and progression tracking.

Literature Review

Title	Author	Methodology	Gaps
A Deep Learning Approach for Automatic Detection of Kidney Diseases using CT Images	Srinivas Aluvala, Pulluri Srinivas Rao	Used CNN-based Deep Learning for detecting kidney diseases, including PKD patterns, from CT scans.	Not PKD-specific; poor detection of small cysts; no integration with genetic data.
Deployed Deep Learning Kidney Segmentation for PKD MRI	Akshay Goel, Gorge Shih	Applied U-Net with EfficientNet Encoder and Transfer Learning for kidney segmentation.	Only single-time MRI analysis; segmentation errors for complex cysts; no growth prediction.
AI Models for Automated Segmentation of Engineered PKD Tubules	Simone Monaco, Nicole Bussola,	Compared U-Net, UNet++, HardNet-MSEG, PraNet, and UACANet with data augmentation and post-processing.	Poor detection of small cysts; used single-donor dataset; requires manual post-processing.
Leveraging Explainable AI (XAI) for Expert Systems in	Latifa Dwiyantri, Hidetaka Nambo	Used RF, LR, SVM, LightGBM, XGBoost, DNN with SMOTE/ROS and XAI tools (LIME/SHAP) for explainable predictions.	Data imbalance; demographic bias; limited real-world validation;

Deep Learning-Based Automated Kidney and Cyst Segmentation Using Multi-Institutional Data	Emma K. Schmidt, Chetana Krishnan , Ezinwanne Onuoha	Used CNN-based segmentation with single vs. multi-institutional training datasets.	Institutional variability; limited validation; focused only on ADPKD cases.
Automatic Segmentation of Kidneys Using Deep Learning for Total Kidney Volume Quantification	Kanishka Sharma, Christian Rupprecht	Used Deep Learning segmentation to calculate Total Kidney Volume (TKV) from MRI scans.	Old dataset; single-modality imaging; lacks modern explainable AI and genetic integration.
Artificial Intelligence in Nephrology: Clinical Applications and	Prabhat Singh, Lokesh Goyal	Reviewed AI and ML models in nephrology, covering PKD segmentation, diagnosis, and prediction.	Broad coverage; not specific to PKD; highlights ethical and interpretability concerns.

Analysis of Literature Survey

Deep Learning Techniques: Most studies used **CNN-based and U-Net models** for kidney and cyst segmentation, showing high accuracy but limited generalization.

Explainable AI (XAI): Recent research started using **SHAP** and **LIME** to make AI models more **interpretable and trustworthy** for clinicians.

Data Limitations: Many studies used **small or single-center datasets**, which cause **bias** and reduce the **model's performance** on real-world data.

Lack of Multi-Modal Systems: Existing works focus on **only imaging or gene data**, missing the benefit of combining **imaging, genetic, and clinical features**.

Need for Long-Term Prediction: Most models analyze data from **a single time point**; there is a strong need for **longitudinal or time-series analysis** to track disease progression.

Research Gap

1. Limited Early Detection Models:

Most existing studies focus on detecting PKD at later stages. There is a lack of **machine learning models** that can **identify early signs** of the disease before severe kidney damage occurs.

2. Lack of Explainability and Trust:

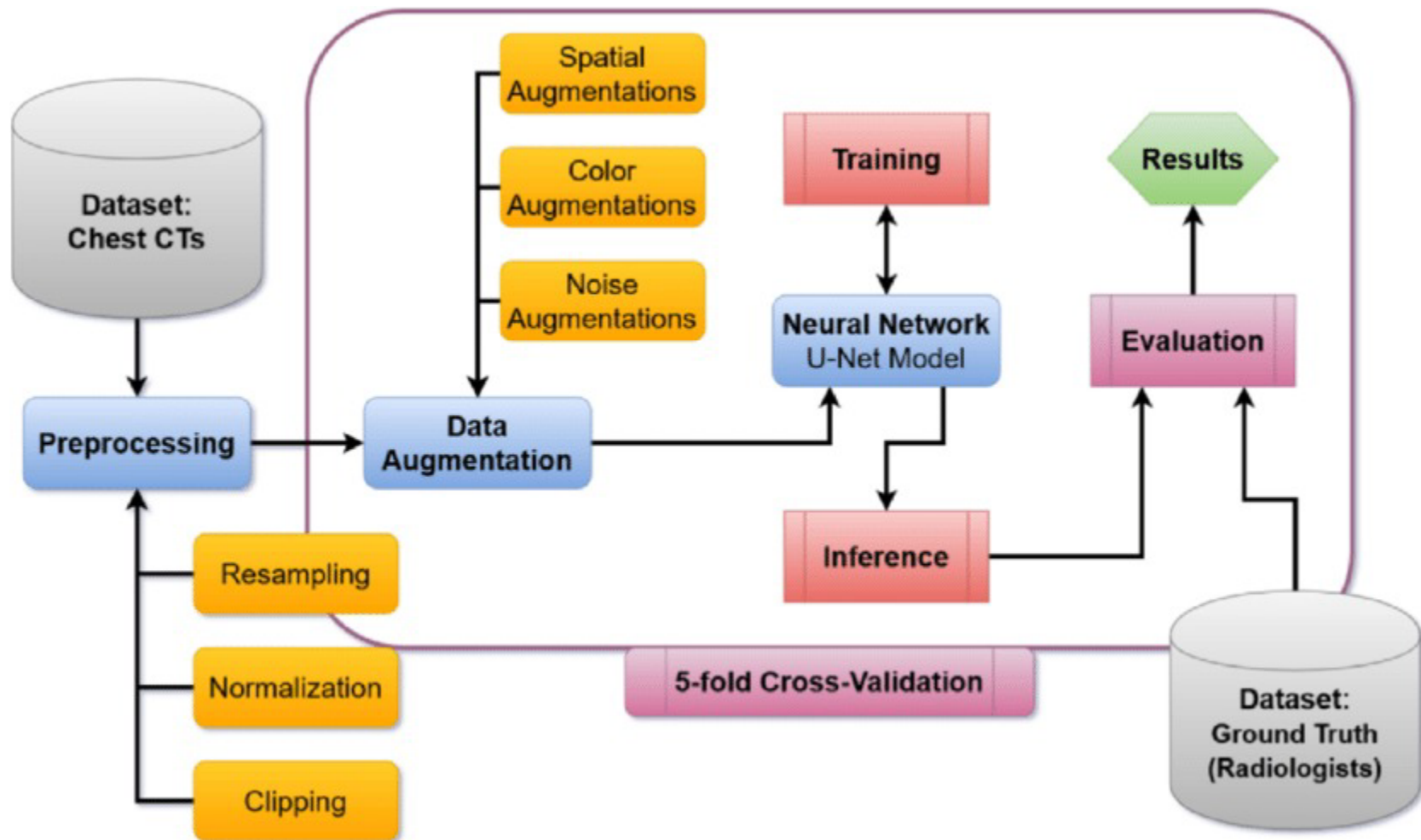
Many deep learning models work like “**black boxes**”, providing results without explaining how decisions are made. This reduces **trust among doctors** and **limits clinical adoption**.

3. Insufficient Multi-Modal Integration:

Current research often uses **only image data (MRI/CT)** or **only genetic data**. There is a **research gap in combining both imaging and clinical/genetic information** to make the diagnosis **more accurate and personalized**.

Problem Definition

Polycystic Kidney Disease (PKD) diagnosis currently depends on manual image analysis and genetic testing, which are time-consuming and often detect the disease at later stages. Existing AI models mainly focus on imaging data and lack early detection capabilities and interpretability. Therefore, there is a need for a machine learning-based system that can automatically detect cysts, assess disease severity, and provide early, accurate, and explainable PKD diagnosis to support clinical decision-making.



Mathematical Foundations of Model

Dice Coefficient for Segmentation

Where TP = true positives, FP = false positives, FN = false negatives.

U-Net for Imaging Segmentation

Objective: Minimize loss for accurate cyst boundaries.

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN}$$

U-Net Equation:

- **Encoder:** Convolutional layers with MaxPooling.
- **Decoder:** Up-Convolution with concatenation.

$$Output = f(\text{Encoder-Decoder with skip connections})$$

How It Works:

1. **Input:** MRI/CT images.
2. **Feature Maps:** Extract hierarchical features.
3. **Output:** Segmented mask for cysts/kidneys